MSG No. 27.82 P_cc2 [Type IV, orthorhombic]

Table 1: Wyckoff site: `2a`, site symmetry: `m'm'2`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,7,8]`
2	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[3,4,5,6]`

Table 2: Wyckoff site: `2b`, site symmetry: `m'm'2`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,2,7,8]`
2	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[3,4,5,6]`

Table 3: Wyckoff site: `2c`, site symmetry: `m'm'2`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ z\right]$	`[1,2,7,8]`
2	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[3,4,5,6]`

Table 4: Wyckoff site: `2d`, site symmetry: `m'm'2`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[1,2,7,8]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[3,4,5,6]`

Table 5: Wyckoff site: `4e`, site symmetry: `.m'.`

No.	position	mapping
1	$[x,\ 0,\ z]$	`[1,8]`
2	$[-x,\ 0,\ z]$	`[2,7]`
3	$\left[-x,\ 0,\ z+\frac{1}{2}\right]$	`[3,6]`
4	$\left[x,\ 0,\ z+\frac{1}{2}\right]$	`[4,5]`

Table 6: Wyckoff site: `4f`, site symmetry: `.m'.`

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ z\right]$	`[1,8]`
2	$\left[-x,\ \frac{1}{2},\ z\right]$	`[2,7]`
3	$\left[-x,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[3,6]`
4	$\left[x,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[4,5]`

Table 7: Wyckoff site: `4g`, site symmetry: `m'..`

No.	position	mapping
1	$[0,\ y,\ z]$	`[1,7]`
2	$[0,\ -y,\ z]$	`[2,8]`
3	$\left[0,\ y,\ z+\frac{1}{2}\right]$	`[3,5]`
4	$\left[0,\ -y,\ z+\frac{1}{2}\right]$	`[4,6]`

Table 8: Wyckoff site: `4h`, site symmetry: `m'..`

No.	position	mapping
1	$\left[\frac{1}{2},\ y,\ z\right]$	`[1,7]`
2	$\left[\frac{1}{2},\ -y,\ z\right]$	`[2,8]`
3	$\left[\frac{1}{2},\ y,\ z+\frac{1}{2}\right]$	`[3,5]`
4	$\left[\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$	`[4,6]`

Table 9: Wyckoff site: `8i`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[-x,\ -y,\ z]$	`[2]`
3	$\left[-x,\ y,\ z+\frac{1}{2}\right]$	`[3]`
4	$\left[x,\ -y,\ z+\frac{1}{2}\right]$	`[4]`
5	$\left[x,\ y,\ z+\frac{1}{2}\right]$	`[5]`
6	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	`[6]`
7	$[-x,\ y,\ z]$	`[7]`
8	$[x,\ -y,\ z]$	`[8]`